

# Anthropomorphism of Virtual Agents and Human Cognitive Performance in Augmented Reality

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## ABSTRACT

In this preliminary study, we aimed to explore the role of anthropomorphism on cognitive task performance in augmented reality (AR). We implemented five different levels of anthropomorphic representations of a virtual agent, ranging from voice to whole-body. Participants performed anagram tasks in the presence of each representation. The results suggest that, except for the voice condition, social presence and cognitive performance increase as the level of anthropomorphism increases.

## 1 INTRODUCTION

Various studies have explored the possibility of observing psychological effects in the presence of virtual agents [2–4, 10, 13, 14, 21, 25, 26]. For instance, it has been shown that the presence of virtual agents may induce social anxiety in some individuals [13] or when treated as an empathic therapist, a virtual agent may help individuals to engage in intimate conversations [4]. Some studies have also shown that the presence of a virtual agent may increase or decrease cognitive performance of individuals [10].

Virtual agents in these studies come in different designs and various visual representations, ranging from no visual representation and voice-only agent [7] to a realistic humanoid agent [26]. Some studies have suggested that different representations may actually have an effect on perceived psychological effects of the agents. For instance, Nowak and Biocca [17] varied the level of anthropomorphism of virtual agents and showed that humans show different social responses to each variation. Yoon et al. [26] proposed that a realistic whole-body agent is preferred over an upper-body or cartoon style agents for remote collaborations. Also, Kim et al. [7] showed that when a virtual agent has a human body representation and shows social behaviors, users feel more confident in the agent's ability to perform tasks compared to a voice-only agent.

However, the question of whether different levels of anthropomorphism play a role in cognitive performance in the presence of a virtual agent is still open. Thus, in this preliminary study, we aimed at exploring possible answers to this question. Thus, we implemented five representations of a virtual agent: voice-only, lips-only, head-only, upper-body and whole-body. Participants performed eight anagrams in the presence of each of these representations. The

preliminary results are presented and discussed in the rest of this paper.

## 2 RELATED WORKS

In 1994, psychologists Nass et al. [16] stated that the interactions of humans with computers are inherently social. Through analyzing five studies in the field of human-computer-interaction, they came to the conclusion that humans do apply social rules when interacting with a computer. They also concluded that humans do so even though they know that it is not necessary.

Since these observations were made when analyzing interactions with non human-like computers, it is not a surprise that people also maintain social rules in the presence of more human-like virtual agents. Bailenson et al. [2], for example, showed that people in virtual reality (VR) keep a larger distance to virtual agents than to virtual objects.

Not only do humans maintain social rules when interacting with virtual agents but Wagner et al. [24] showed that humans also expect virtual agents to maintain social rules. They employed an agent to guide people through an art gallery. When evaluating participants' opinions on the virtual agent, they found that some participants felt insulted when the agent violated social norms.

From less human-like agents like computers to virtual representations of humans, virtual agents consistently appear to feature some sort of social presence. The degree of social presence, however, can vary. Moon et al. [11] conducted a study examining the difference in experience when shopping with the virtual avatar of another person compared to just hearing the voice of the shopping companion. They found that social presence was perceived as intensified if the shopping partner was represented by a virtual avatar.

Oh et al. [18] created a systematic review of social presence. In their work, they stated that one of the features determining social presence is the visual representation of the communication partner. On one hand, this includes the actual presence or absence of a visual communication partner. On the other hand, visual representation also encompasses, according to Oh et al., the degree of visual realism of such an associate. In their definition visual realism is made up of photographic, anthropomorphic and behavioral realism. In this context, photographic realism determines how realistic something appears. Anthropomorphic realism describes to which extent something looks like a human. And behavioral realism covers the degree of similarity to actual human behavior.

Supporting Oh et al.'s review, Steptoe et al. [22] were able to demonstrate that flaws in photographic realism will impair the perceived social presence of virtual agents in AR. Visual differences between virtual content and real world reduce social presence. Having virtual and real content next to each other, seems to highlight differences, lowering photographic realism and therefore social presence.

In an attempt to heighten the level of social presence, Kim et al. [8] tried to merge virtual content with the real world. They did this by having a fan blowing in the room that affected both, the virtual content as well as the real world. They observed that when

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the virtual agent reacted to the gust of wind, so behaving more realistically, the level of social presence was elevated.

Several studies showed that even small efforts in order to increase behavioral realism in virtual agents can greatly impact the perception of their social presence. For example, in the previously mentioned study of Bailenson et al., humans kept even larger distance to an virtual agent when they had eye contact with it [2].

But humans do not only perceive the degree of social presence of virtual agents, they also evaluate them socially and derive consequences for themselves. In 2011, Guadagno et al. [5] observed that participants' social evaluation was impacted by the facial expression of the agent provided, e.g. a smiling agent compared to an agent displaying a serious facial expression. However, this social evaluation of virtual agents can have positive as well as negative implications.

A positive consequence can be found in the work of Nass et al. [15]. They observed that humans are polite to computers, even in a text-based scenario. And Lester et al. [9] found in one of their studies that school children perceived a virtual agent (in this case an animated bug) as helpful.

However negative consequences can appear as well. Pertraub et al. [20] tried in their work to evaluate whether virtual agents could induce social anxiety. They had study participants make a speech in a virtual environment in front of a virtual audience. As hypothesized, they observed that the participants indeed felt anxious, especially when the virtual audience behaved unappreciative.

Furthermore, Ye Pan and Anthony Steed have run some experiments on how humans perceive agents represented in different ways. For instance, in one study [19] they found out that participants preferred to seek advice from expert representations with a tendency for seeking advice from a robot or a video of a real human. In contrast, receiving advice from a virtual humanoid agent was rarely preferred as synthetic visual cues may have prevented trust building and resulted in a negative bias.

Studies have also shown that similar to performing tasks in the presence of real humans, presence of virtual agents may affect task performance. An improvement of task performance is known as social facilitation and a deterioration of the performance is known as social inhibition [27]. Miller et al. [10], for instance, showed the occurrence of social facilitation and inhibition effects in an augmented reality (AR) environment, as participants performed better in an easy task and worse in a hard task in the presence of a virtual agent.

Therefore, in this work, we aimed to assess the social presence felt by the individuals towards different levels of anthropomorphic representations of a virtual agent. In addition, we were interested to know whether this social presence may affect participants' cognitive performance. Thus, each participant was given eight anagram tasks to solve in the presence of each representation.

### 3 METHODS

#### 3.1 Materials

##### 3.1.1 Cognitive Task

Participants solved an anagram task similar to the study by Miller et al. [10]. A pool of German-language anagrams was compiled. These anagrams have been used and tested in previous studies [6, 23]. In each condition, eight anagrams in a randomized order were presented to the participants.

##### 3.1.2 Levels of Anthropomorphism

The degree of anthropomorphism was varied through five different representations of the virtual agent: voice, lips, head, upper-body, and whole-body (Figure 1). The instructions on how to perform the cognitive task and navigate to the next question was given by each representation. In case of a visual representation, the lips of the agent was synced with the given auditory instructions. Also, the whole-body and upper-body representations had idle standing

animations. Finally, each representation was placed above a virtual Amazon Echo.

##### 3.1.3 Virtual Amazon Echo

The virtual Amazon Echo was modeled in Blender and animated with a blue light so that when the virtual agent spoke, the Echo lit up. This representation of the Amazon Echo was intended to facilitate understanding of otherwise unfamiliar conditions (e.g., lips only condition) by creating a plausible and possibly familiar context (i.e., obtaining assistance from a real Amazon Echo). In addition, the echo was placed on a virtual table.

##### 3.1.4 Augmented Reality Display

To display our AR content, we used Nreal, which is a light weight head-mounted display (HMD) and looks more like usual every-day glasses. In addition, it comes with a set of correcting lenses to be magnetically attached to the glasses and different shapes of nose pads. The glasses are connected via cable to a computing unit which can be attached to the belt or put in the pocket. The computing unit has an Android operating system. In addition, Nreal has a hand-held controller with three buttons. The intuitive mapping entails minimal learning, which is a major advantage for participants with no prior experience with AR.

##### 3.1.5 Questionnaire

To measure social presence, we employed the social presence questionnaire by Bailenson et al. [1]. It has a total of six statements which can be evaluated on a range of -3: "I strongly disagree" to 3: "I strongly agree.". This questionnaire was administered after each experimental condition.

### 3.2 Procedure

Due to the COVID-19 pandemic restrictions, we performed our study with only two participants. They were both female and aged 19 and 48 years old from the same household as the experimenter (a given restriction for conducting empirical studies at the time). None of them had prior experience with AR, but one had experience with VR.

After a short briefing on how to use the controller, participants were asked to wear the AR HMD, while holding the computing unit in their non-dominant hand and the controller in the dominant hand. Thereafter, the program could be started by the participant to work through the tutorial with the help of the experimenter. After this learning phase, the experimenter left the room to enable the participant to carry out the experiment alone.

After wearing the AR HMD, a virtual table with the virtual Amazon Echo on top was visible for the participant in the AR environment. Right behind the table, a virtual canvas could be found, on which the anagrams were displayed. Buttons to get to the next anagram were shown below the canvas. Each condition was presented to the participants by showing the five representations of the virtual agent in a randomized order. Meanwhile eight anagrams were displayed on the canvas one after another. The performance was measured by recording the time taken to answer each anagram. The questionnaire was also administered after each condition.

### 4 RESULTS

To evaluate the cognitive task performance, the number of correct responses and the response time per response were measured. Figure 2 shows the percentage of correct answers per condition. It can be seen that the whole-body condition has the highest correct answers. Except for the voice condition, the percentage of correct responses decreases as the degree of anthropomorphism decreases.

Figure 3 shows the response time per correct response. The graph shows a faster response time for the whole-body ( $M = 6.457$ ;  $SD = 3.161$ ) and the upper-body ( $M = 5.297$ ;  $SD = 3.565$ ) conditions.

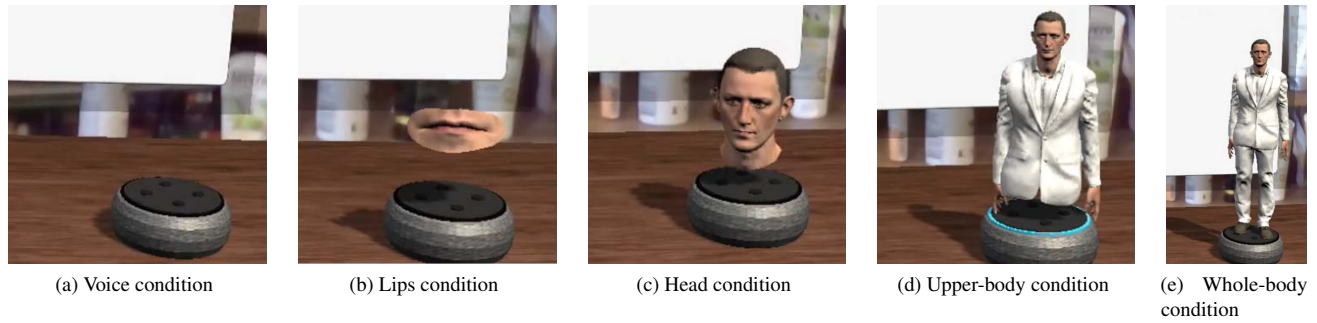


Figure 1: Different levels of anthropomorphism of the virtual agent: the first condition is voice, represented by the Amazon Echo. The other four conditions: lips, head, upper-body, and whole-body, are projected above a virtual Amazon Echo.

Again, the participants gave the correct answers faster in the voice condition ( $M = 6.202$ ;  $SD = 1.672$ ) compared to the head and lips conditions.

The results of the social presence questionnaire show that the whole-body condition had an average value of 2.5, indicating that the participants perceived only this condition as socially present in the physical room. The other conditions showed a negative value for social presence which indicates an unperceived social presence.

In addition, the result of a Pearson correlation with  $r = -.3$  and  $p = .032$  suggests a negative correlation between the social presence score and the required response time for giving a correct answer in the cognitive task. That is, the higher the perceived social presence (which may arise from the degree of anthropomorphism), the shorter the response time required for a correct response.

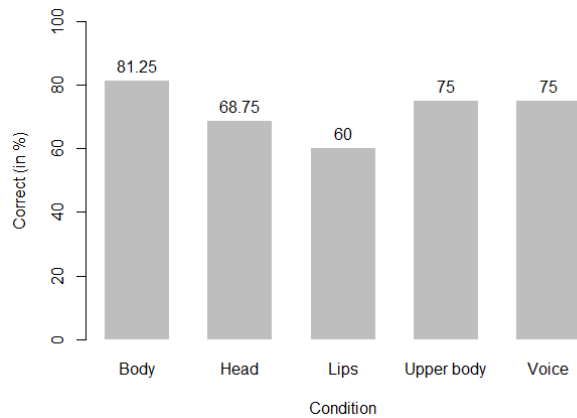


Figure 2: Percentage of the correct answers

## 5 DISCUSSION AND CONCLUSION

The presented exploratory results suggest that the degree of anthropomorphic representation of a virtual agent may have an impact on the cognitive task performance in AR environments. Here, we observed that the higher the level of anthropomorphism, the better the cognitive performance. Moreover, the perceived social presence was higher for more anthropomorphic agents. Thus, social presence may play the role of a mediator here. Meaning that a better performance in cognitive tasks may be due to the reason that the more anthropomorphic representations of the virtual agent induce a higher sense of being socially present with the participant in the room. Thus, we could observe a tendency towards social facilitation

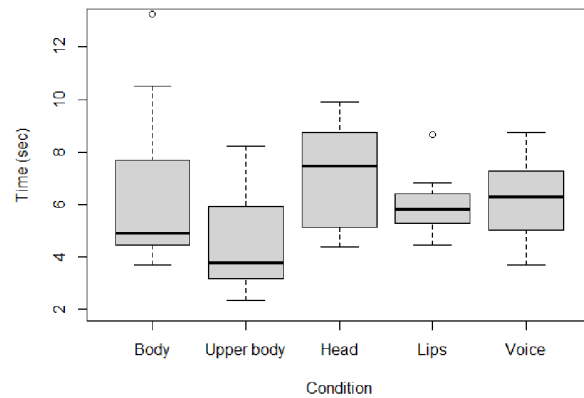


Figure 3: Response time needed for the correct answers

by more anthropomorphic agents.

However, the voice-only agent did not follow this pattern and showed the best cognitive performance right after the whole-body and upper-body representations. A reason for this could be the familiarity of the participants with today's common voice assistant systems. On the other hand, the low perceived social presence and cognitive performance observed by performing in the presence of the lowest anthropomorphic representations (i.e., lips-only and head-only) could be explained by the Uncanny Valley concept [12]. It suggests the evocation of unpleasant emotional responses due to an inhuman agent. The voice condition, however, may not be affected by this concept, since it may resemble a familiar scenario of interaction with today's voice assistant.

A limitation of our study is that to evaluate the cognitive task responses, we recorded the voice of the participants and analyzed them manually. However, an automatic evaluation of the responses by a speech recognition system could be implemented in future studies. Another possibility is to expand the range of functions of the controller. For example, moving the letters using a drag and drop function would be conceivable for solving the anagrams. However, individual differences in using 3D input devices such as an AR controller, may induce additional cognitive load for participants and affect their cognitive performance. An investigation into the extent to which this concern may arise is appropriate and should be undertaken prior to conducting future studies.

In the end, the biggest limitation of this study was its small number of participants. Thus, we were able to report only our exploratory results. However, with a larger number of participants in future studies, one may carry out a valid inferential evaluation for hypothesis testing.

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